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Problem

- HDRIs are standard light sources for image-based lighting, but casual LDR panoramas **clip at the saturated area**.
- Clipped highlights destroy **diffuse** irradiance and **specular** reflections.
- Pixel losses treat all errors similarly, even though rendering quality is dominated by lighting integrals.

Related Work

- Single-image HDR reconstruction improves local pixels, but does not directly supervise IBL behavior.
- DiffHDR shows the value of video diffusion priors for exposure recovery.

Missing: losses aligned with how HDRIs are rendered.

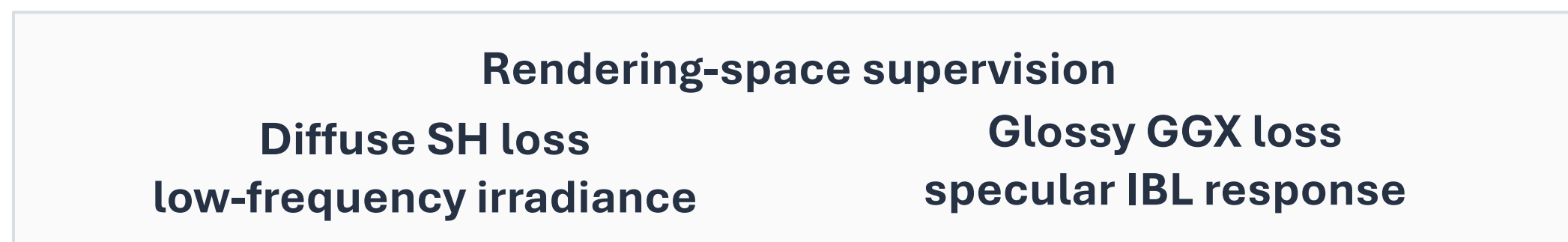
Our Approach

- Fine-tune Wan 2.1 VACE 14B with LoRA for clipped-HDRI promotion.
- Use mask-conditioned inpainting of saturated highlight regions.
- Add diffuse SH and glossy GGX rendering-space losses for IBL fidelity.
- Train in linear radiance with log encoding for VAE compatibility.

References

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Method



$$\mathcal{L}_{\text{diffuse}} = \frac{1}{27} \sum_{c,k} (\log(1 + c_k^{\text{pred}}) - \log(1 + c_k^{\text{gt}}))^2 \quad \mathcal{L}_{\text{glossy}} = \|\log(1 + \mathbf{E}_{\text{glossy}}^{\text{pred}}) - \log(1 + \mathbf{E}_{\text{glossy}}^{\text{gt}})\|_2^2$$

$$\mathcal{L} = \mathcal{L}_{\text{noise}} + \lambda_d \mathcal{L}_{\text{diffuse}} + \lambda_g \mathcal{L}_{\text{glossy}}, \quad \lambda_d = \lambda_g = 0.05$$

- The LDR panorama is encoded as a single-frame video input. Clipped regions are mask-conditioned for inpainting.
- Diffuse SH compares 27 second-order coefficients in log space, directly supervising irradiance.
- Glossy GGX compares prefiltered specular lobes over 64×128 HDRI maps.

Results

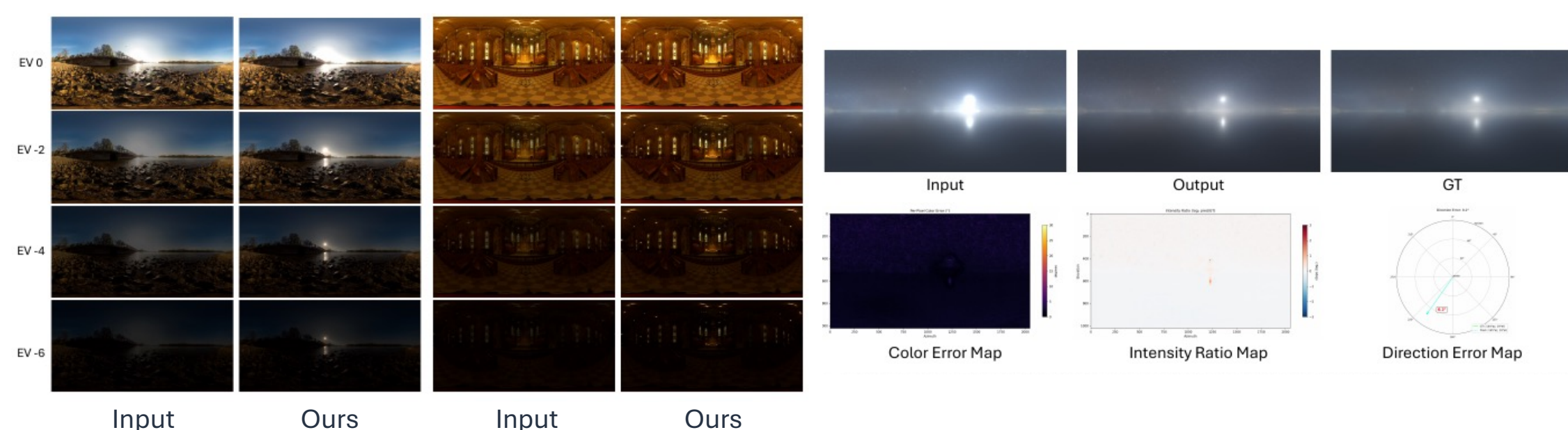


Fig.1 The model recovers clipped highlight details from LDR panorama input and achieve low color error, direction error, and close intensity ratio with the Ground Truth HDRI.

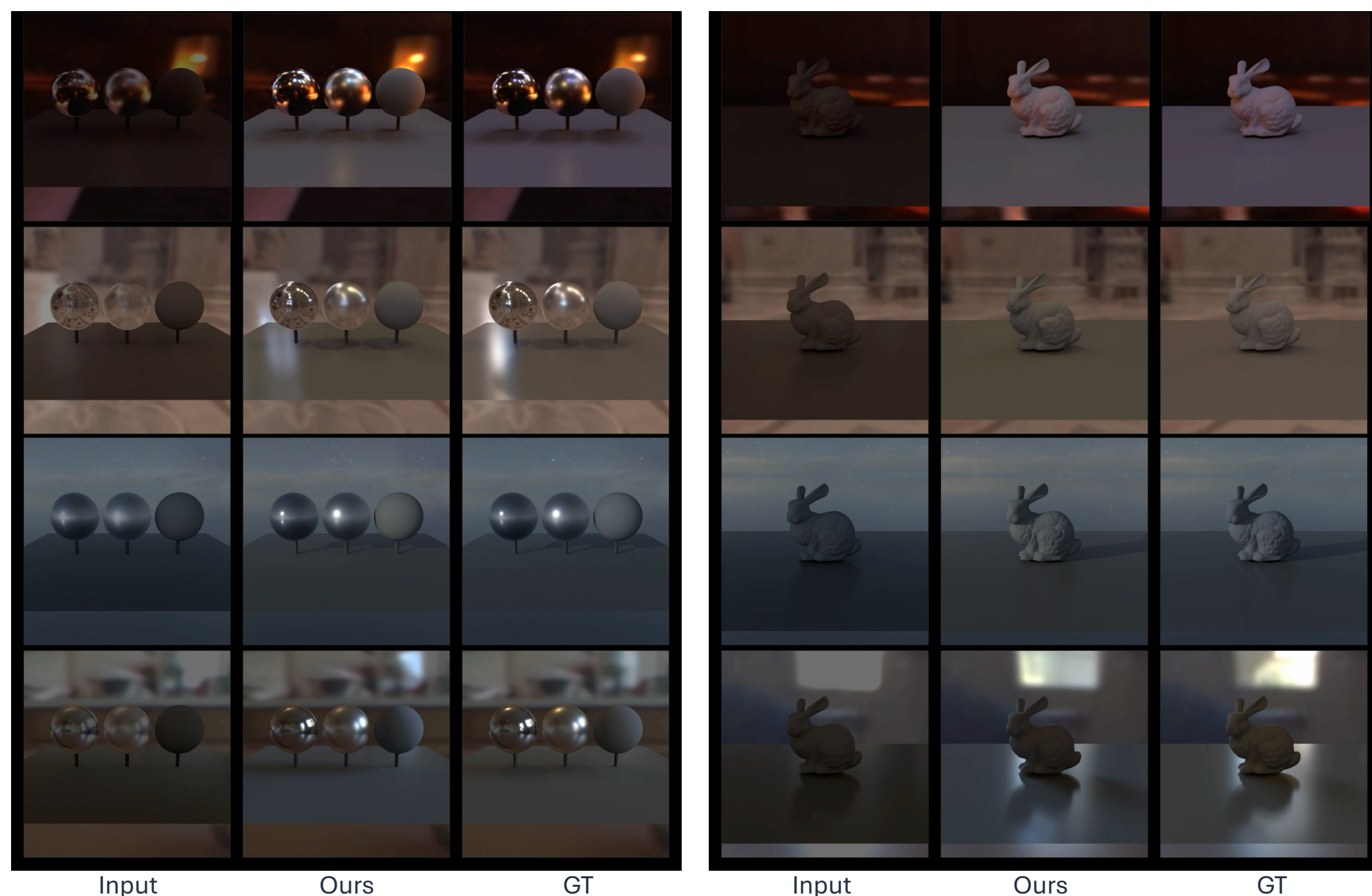


Fig. 3. Relighting comparison. Our reconstruction recovers the illumination intensity, color, shadowing, and specular reflections needed for realistic relighting.